



Premier University Chittagong

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Assignment

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Assignment Name : A day in the life of a HTTP request

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A Day in the Life of an HTTP Request/Reply

Introduction

When we open a website in a browser, many networking protocols and devices work together to complete the communication. This complete journey is called “A Day in the Life of an HTTP Request/Reply.” It explains how data travels from a user’s computer to a web server and back again.

According to the networking model, communication happens through multiple layers such as application layer, transport layer, network layer, and link layer.

Step 1: User Enters a URL

Suppose a user types “<http://www.example.com>” into a browser.

The browser first checks whether the IP address of the website is already stored in cache. If not, it needs to find the server’s IP address using DNS (Domain Name System).

Step 2: DNS Query and Reply

DNS translates a domain name into an IP address.

For example, DNS may translate www.example.com into the IP address 93.184.216.34.

The browser sends a DNS request to a DNS server. After searching its records, the DNS server replies with the IP address of the web server.

Now the browser knows where to send the HTTP request.

Step 3: ARP Resolves MAC Address

Before sending data on the local network, the host needs the MAC address of the destination device (usually the router/default gateway).

ARP (Address Resolution Protocol) is used for this purpose.

The sender broadcasts an ARP request asking, “Who has 192.168.1.1?”

The router replies with its MAC address.

The router then replies with its MAC address.

Now the sender can create an Ethernet frame.

Step 4: TCP Connection Establishment

HTTP usually uses TCP as its transport protocol.

Before data transfer, TCP performs a 3-way handshake:

1. SYN

The client sends a SYN message to the server requesting a connection.

2. SYN-ACK

The server responds with a SYN-ACK message acknowledging the request.

3. ACK

Finally, the client sends an ACK message back to the server.

Now the TCP connection is established.

Step 5: Browser Sends HTTP Request

The browser sends an HTTP request message.

For example, the browser may send a GET request asking for a webpage from the server.

This message belongs to the application layer.

Step 6: Encapsulation Through Layers

The HTTP message moves downward through networking layers.

Application Layer

Creates HTTP request.

Transport Layer

TCP adds:

- Source port
- Destination port
- Sequence number
- Acknowledgment number

Network Layer

IP adds:

- Source IP address
- Destination IP address

Link Layer

Ethernet adds:

- Source MAC address
- Destination MAC address
- Error detection bits

Frames encapsulate datagrams at the link layer.

Step 7: Data Travels Through Routers

Routers examine the destination IP address and forward packets toward the destination network.

At every hop:

1. Old Ethernet frame removed
2. New frame created
3. Packet forwarded to next router

The IP packet usually remains unchanged except for fields like TTL.

Step 8: Web Server Receives Request

The server receives the frame and removes headers layer by layer:

1. Ethernet header removed
2. IP header removed
3. TCP segment extracted
4. HTTP request delivered to web server software

The server processes the request and prepares the response.

The server then creates an HTTP response such as “200 OK” and sends the requested webpage data back to the client.

Then the HTML page data is attached.

Step 9: HTTP Reply Sent Back

The server sends the response back using the same layered process:

1. HTTP response created
2. TCP segment formed
3. IP datagram created
4. Ethernet frame transmitted

Routers again forward the packets through the Internet until they reach the client.

Step 10: Browser Displays the Webpage

The client receives the packets, reassembles them in order using TCP, and passes the HTTP response to the browser.

The browser:

- Reads the HTML
- Downloads images, CSS, and JavaScript if needed
- Renders the webpage for the user

Finally, the webpage appears on the screen.

Conclusion

“A Day in the Life of an HTTP Request/Reply” demonstrates how different networking protocols cooperate to load a webpage successfully. DNS resolves names, ARP resolves MAC addresses, TCP ensures reliable communication, IP handles routing, and HTTP transfers the actual web content. Together these protocols enable fast and reliable web communication across the Internet.